

The scripts dir:

```
(basic_env) (epsilon_env) jali13_c@Apus:~/PhD/EPSILON/DEEPLARNING/mask_new/structured_software_steps$ ls -lrt scripts/
total 64
-rwxr-xr-x 1 jali13_c jali13_c 183 Aug 25 15:35 make_bed_parallel_config.yaml
-rwxr-xr-x 1 jali13_c jali13_c 2645 Aug 25 15:37 make_bed_parallel.smk
-rwxrwxr-x 1 jali13_c jali13_c 442 Aug 25 15:44 subsample_loci.R
-rwxrwxr-x 1 jali13_c jali13_c 189 Aug 25 15:46 remove_callable_loci.sh
-rwxrwxr-x 1 jali13_c jali13_c 9453 Aug 25 15:51 model_definitions.py
-rwxrwxr-x 1 jali13_c jali13_c 9817 Aug 25 15:54 train.py
-rwxrwxr-x 1 jali13_c jali13_c 2264 Aug 25 15:58 extract_features.py
-rwxrwxr-x 1 jali13_c jali13_c 2264 Aug 25 15:59 extract_features_call.py
-rwxrwxr-x 1 jali13_c jali13_c 5106 Aug 25 15:59 extractor_functions.py
-rwxrwxr-x 1 jali13_c jali13_c 4986 Aug 25 16:15 call_tmp.py
```

```
reference_genome: "hs37d5.fa"
reference_genome_fai : "hs37d5.fa.fai"
bamfile: "sample1.bam"
chromosome: "1"
chunk_size: 5000000
min_coverage: 30
output: "merged.processed.mpileup.txt"
```

make raw bed

```
#!/usr/bin/bash
```

```
snakemake --snakefile scripts/make_bed_parallel.smk --configfile scripts/make_bed_parallel_config.yaml
```

(optional) subsample loci bed

OUTPUT

```
raw_bed="merged.processed.mpileup.txt"
output_subsampled_bed="subsampled.mpileup.txt"
subsampling_strength=0.1

Rscript scripts/subsample_loci.R $raw_bed $output_subsampled_bed $subsampling_strength
```

remove loci that will be called

```
callable_loci_bed="callable.bed"
input_bed="subsampled.mpileup.txt"
output_bed="subsampled.mpileup.final.txt"

scripts/remove_callable_loci.sh $callable_loci_bed $input_bed $output_bed
```

extract features for training

```
nranks="8"

mpirun -np $nranks python3 scripts/extract_features.py \
--bed loci.bed \
--reference-genome reference.fa \
--bam reads.bam \
--output output_dir \
--genomic-window 100
```

train the model

```
training_data_dir="npz_output_test"

python3 scripts/train.py \
--data_dir "$training_data_dir" \
--max_epochs 1000 \
--zero_keep_prop_multiplier 10.0 \
--optimizer adam \
--learning_rate 1e-4 \
--patience 3 \
--max_error_rate 0.05 \
--output_weights fitted.weights.h5
```

extract features for the target tumor sample

```
nranks="8"

mpirun -np $nranks python3 scripts/extract_features_call.py \
--bed calling_region.bed \
--reference-genome reference.fa \
--bam tumor.bam \
--output call_data_output_dir \
--genomic-window 100
```

call variants (temporary)

```
data_call="calling_data_dir"
error_model="trained_model.weights.h5"

python3 scripts/call_tmp.py --calling_data_dir $data_call --error_model $error_model
```

